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Intelligent detection of deterioration in cultural stone heritage

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ABSTRACT

Vision-based periodic examination of the deterioration of stone monuments over time is labour and time intensive. Especially, in cases involving large-scale immovable cultural heritage, the workforce is considerably increased, along with the possibility of occurrence of errors. Any misdiagnoses in the deterioration may cause irreversible structural problems in monuments, and thus, it is necessary to develop alternative examination methods. Computer-vision methods represent an effective solution to eliminate both human errors and difficulties in the field. Therefore, this study aims to adopt the Mask R–CNN algorithm, which is a computer-vision method, to detect and map the deteriorations observed in the Gümüşler archaeological site and monastery (cracks, discontinuities, contour scaling, missing parts, biological colonization, presence of higher plants, deposits, efflorescence, and loss of fresco). First, 1740 images were collected from the site, and the model was trained by labelling the distortions in these images according to their types. Later, the model was tested on four outdoor and two indoor views. The developed model achieved an average precision ranging between 91.591% and 100%, and the mean average precision was 98.186%. These results demonstrated that the proposed algorithm can enable mapping to promptly and automatically detect the deterioration in large monuments.

1. Introduction

Atmospheric processes cause deterioration in cultural heritage monuments. Restoration works are very important in order to protect these monuments and to be passed on to future generations. For these studies, the preparation of degradation maps is essential in order to understand the development and origins of the destruction. In addition to the traditional method based on visual examinations in the literature, digital tools, which are the products of technological progress, have been used to detect degradation. With terrestrial laser scanning, which is one of the digital tools, extremely sensitive 3D models were created and stone weathering determinations were made [1,2]. However, in cases where the height of the monument is high or direct access is limited, it is very difficult to examine and map the types of deterioration by both methods. In certain studies, drones were adopted to solve the accessibility problems. Owing to the high range and altitude capability of drones, high-resolution 3D models could be obtained, and deterioration analyses were performed using these models [3–6]. Although these methods are valuable in realizing conservation studies, interpreting the data obtained from the field is highly time consuming. Moreover, human errors may occur when analysing the data obtained from drones, which is particularly labour intensive owing to the extremely complex mechanism of the stone deterioration. Recently, artificial intelligence methods, which are widely applied in different fields such as medicine [7], agriculture [8] and engineering [9], can be used to overcome the problems encountered in the study of historical structures. While some researchers prefer machine learning among artificial intelligence methods to examine historical structures [10–13], some researchers have used deep learning (DL) algorithms [14–19]. Bassier et al. [10] determined the building elements (ceilings, floors, roofs, beams, and walls) by using the support vector machine method. Valero et al. [11,12] identified the types of deterioration in the stone wall of the Chapel Royal in Stirling Castle in Scotland. Nazarian et al. [13] performed the damage assessment of a historical building by using machine learning methods. Wang et al. [14] used the faster region with convolutional neural network (R–CNN) algorithm to determine the types of efflorescence and spalling degradation that develop in brick buildings. Wang et al. [15] developed a two-stage model based on Faster R–CNN and Mask R–CNN to detect the damage in historic glazed tiles. Zou et al. [16] developed an artificial intelligence algorithm to enable the control of intact and

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Fig. 1. Some examples of complex structures carved into pyroclastic rocks; a) Kandovan (Iran), b) Varde Monastery (Georgia), c) Cappadocia region (Turkey).

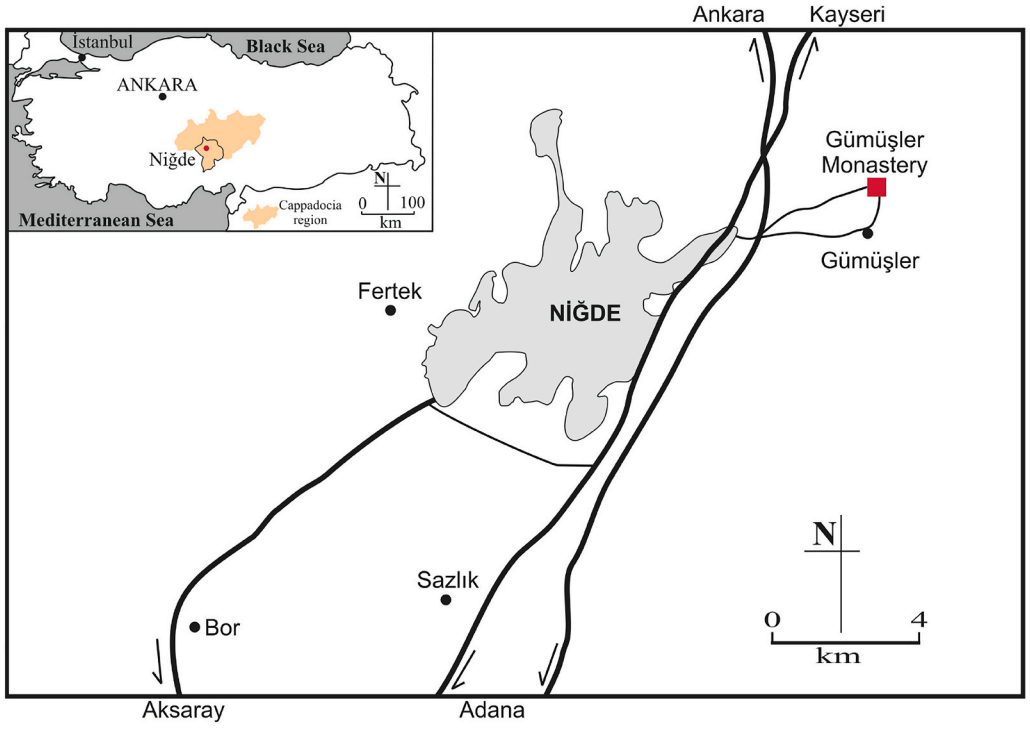

Fig. 2. Location of Gümüşler monastery.

impaired components in a historical structure. Hatır et al. [17] used a DL algorithm to classify the types of deterioration and fresh rock observed in stone monuments and achieved an accuracy rate of as high as 99.4%. Moreover, Hatır and İnce [18] developed a DL algorithm to automatically detect building stone lithologies and evaluated their approach considering the natural stones used in a historical building. Mansuri and Patel [19], who used the Faster R–CNN algorithm to examine the deterioration types of spalling, cracks, and exposed bricks in historical brick structures in India, obtained a mean average precision (mAP) of 0.915.

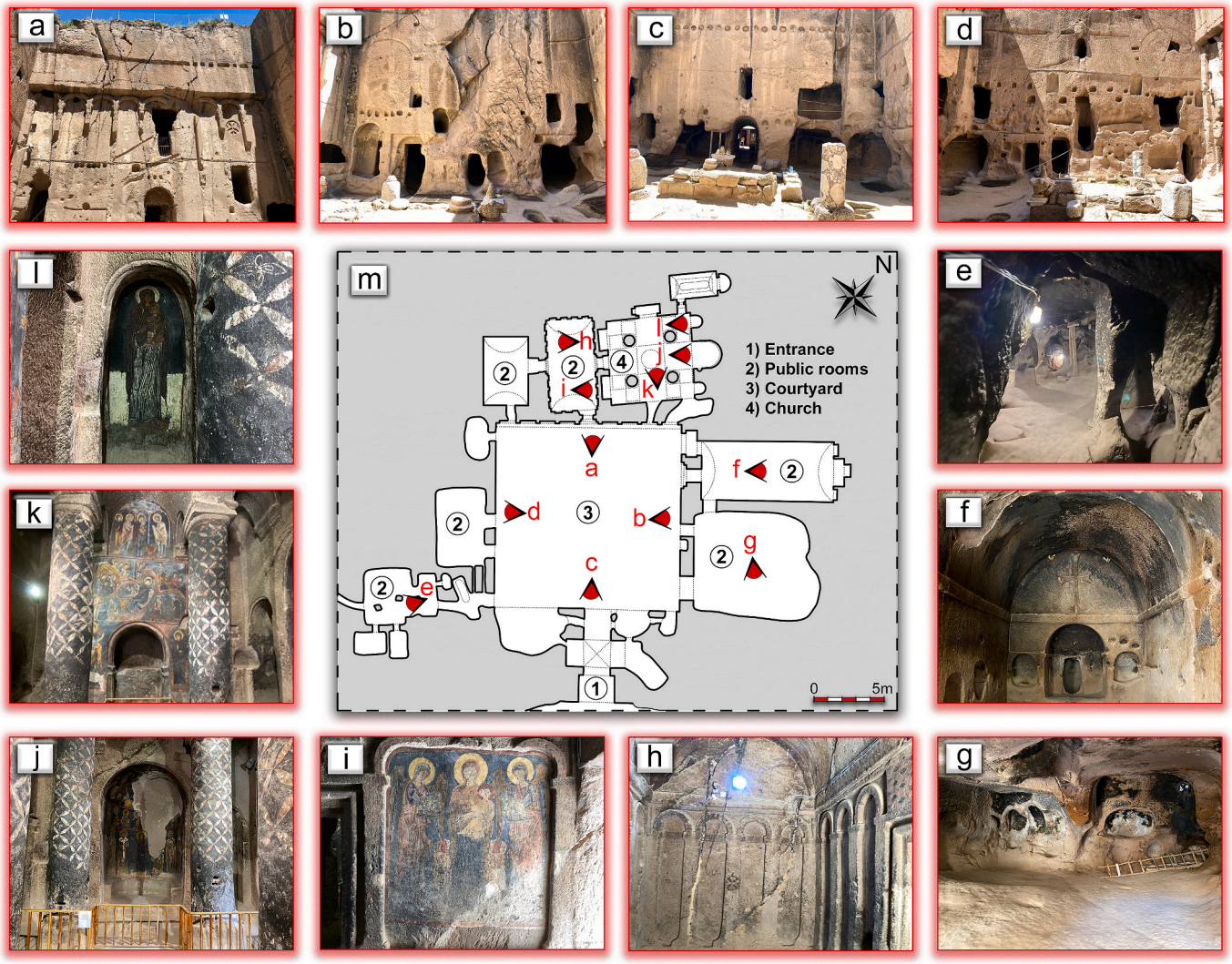

Fig. 3. Gümüşler Monastery interior-exterior views and plan; a-d) courtyard facade views, e) underground city view, f-g) view of public rooms, h) church narthex view, i) fresco on the east wall of the narthex, j) fresco on the apse, k) fresco on the north wall, l) Mary and baby Jesus fresco, m) Monastery plan (rearranged according to Gough [2]).

These studies demonstrate that artificial intelligence algorithms can enable the detection of building materials and the deterioration of these materials. However, the computer-vision algorithms developed to detect deteriorations in cultural heritage sites have been tested only in certain parts of the monuments. Moreover, only classification studies have been performed to examine the stone deterioration in monuments, and the types of deterioration have not been investigated extensively. However, the success of DL algorithms in large-scale cultural heritages exposed to complex types of deterioration has not been tested in previous studies. The success of DL algorithms can be extremely effective in preserving cultural heritage, especially in monuments built on pyroclastic rocks that are spread over large areas, easy to process and sensitive to atmospheric processes.

Throughout history, primarily housing-cave-type structures were built on pyroclastic rocks to protect from against natural disasters and attacks, and in later periods, city complexes in combination with places of worship were established (Fig. 1). A representative example is the Cappadocia region located in the centre of Anatolia, which includes the cities of Aksaray, Nevşehir, and Niğde and involves unique rock structures (Figs. 1c and 2). With the arrival of the Apostle Paul in 53 AD, the foundations of Christianity were laid in this region, and many places of worship such as chapels, churches, and monasteries carved in rock were established in the following periods [20]. A notable monument created for worship in the region is the Gümüşler Monastery, which is carved in a single rock. The monument has undergone considerable deterioration due to the weathering conditions in the region and structural–geological problems of the bedrock. Several researchers have attempted to examine these problems [21–24]. Notably, Gough [21,22] examined the preservation and restoration of the wall frescoes in the monastery. İnce et al. [23] investigated the change in the freeze–thaw cycles with the index and mechanical properties of the rock constituting the monument. Korkanç et al. [24] investigated the engineering geological problems in the monastery and highlighted a basis for conservation planning. However, no paper has reported on the determination and mapping of the types of deterioration observed in this unique monument. Because the height of the building is nearly 14 m, which renders access challenging, it is difficult to examine the types of deterioration.

Therefore, this study was aimed at developing an artificial intelligence algorithm to automatically detect and map the deteriorations across a complex historical building that was spread over a wide area and had a large height. To this end, the Mask R–CNN method was trained on the data pertaining to the Gümüşler Monastery in terms of nine deterioration types (cracks, discontinuities, contour scaling, missing parts, biological colonization, presence of higher plants,

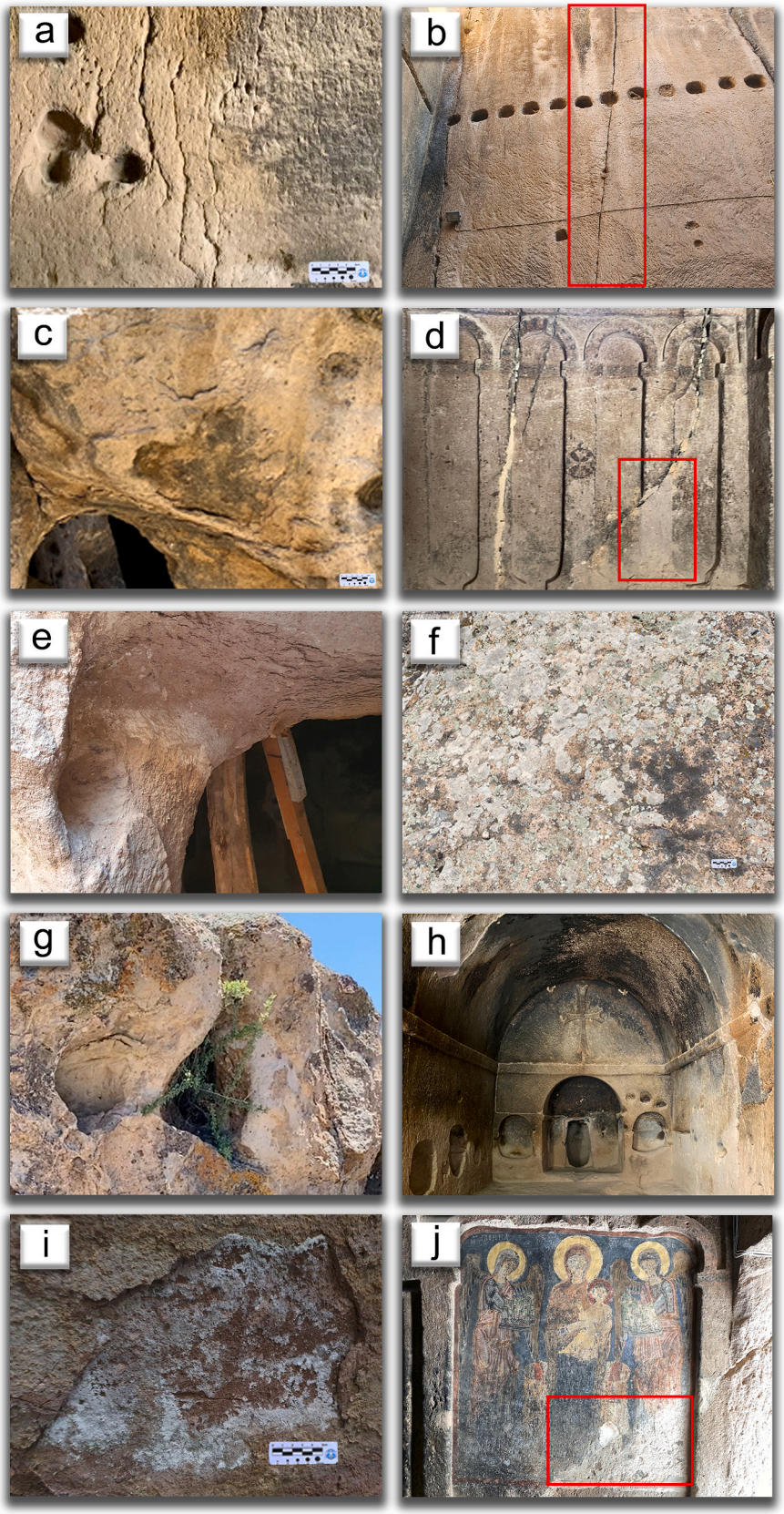

Fig. 4. Types of deterioration observed in Gümüşler Monastery; a) crack, b) discontinuous, c) contour scaling, d) and e) missing part, f) biological colonization, g) higher plant, h) deposit, i) efflorescence, j) loss of fresco.

deposits, efflorescence, and loss of fresco). The proposed algorithm was tested on six images compiled from the courtyard and interiors of the Gümüşler Monastery. The proposed model could realize automatic detection and mapping with a high accuracy for all types of deteriorations, ranging from small to large-scale.

2. Material

2.1. Description of the Gümüşler Monastery

The Gümüşler site and monastery, built in the Byzantine period (7–11th century), is located in the south of the Cappadocia region (Fig. 3). Carved out of a single rock mass, this monument includes a church, various public rooms, a cellar, a kitchen, a burial chamber, and a two-storey underground city. The monastery, which exhibits unique features in the form of icons and complex plan scheme, was included in the UNESCO tentative list in 2012 under the title of "Historical Monuments of Niğde". The entrance to the monument is through a narrow corridor in the southeast direction. On both sides of this corridor are the kitchen, refectory, and various sized stores for wine and oil. This passage space, which is a barrel vault, leads to a square courtyard with dimensions of approximately 13.50 × 13.50 m (Fig. 3a–d). Food stores and graves are present in this courtyard, which is approximately 14 m high. Various spaces and outbuildings (Fig. 3e) are present in the west of this courtyard, and shelters are located in the east (Fig. 3f and g). The church structure is located opposite to the entrance. The narthex of the church is covered with a barrel vault (Fig. 3h), and one wall involves a fresco with the Mary, the child Jesus, Gabriel, and Mikael (Fig. 3i). Four columns exist in the church, which has a closed Greek cross-plan. The side naves are covered with barrel vaults, and the middle nave is covered with a dome. On the main apse of the church is a fresco with Jesus, two angels, symbols of the Biblical authors and the Virgin Mary, and the Apostles (Fig. 3j). On the northern cross arm of the church plan are scenes of the birth of Jesus and his presentation to the temple, the delivery of good news to the Virgin Mary and Baptist John and St. Stephanos (Fig. 3k). To the east of the altar is the smiling Virgin Mary fresco, which is known to be the only one in Anatolia [24] (Fig. 3l). There are two grave niches in the room to the north of the church.

2.2. Geological setting of the Gümüşler Monastery

The underground rock structures in the investigated area were carved in younger ignimbrites overlying the Niğde Group metamorphics. The Niğde Group rocks consist of Gümüşler, Kaleboynu and Aşığediği metamorphics from the bottom to the top, cut by Üçkapılı Granodiorite in certain places [25]. In the study area, the ignimbrites overlying the unconformable Niğde Group metamorphic rocks are known as Kızılkaya ignimbrites. Kızılkaya ignimbrites represent the most widespread tuff unit in the Cappadocian Volcanic Province, in which they form flat surfaces over an area of ~8500–10600 km^2 with a volume of 180 km^3 [26,27]. The framework is incipiently strongly welded with occasional columnar jointing. The K/Ar age dating studies yielded dates between 5.5 and 4.3 Ma for the Kızılkaya ignimbrite [26–29]. Moreover, carved underground rock structures exist, especially in the lower part of the region. The ignimbrites in the lower part of the rock structures exhibit a high porosity, low unit weight, and low resistance and are more sensitive to the atmospheric conditions than the ignimbrite in the upper parts [5].

2.3. Characterization of the deterioration of Gümüşler Monastery

With time, due to atmospheric or anthropogenic effects, partial or significant deterioration of the monument integrity may occur in cultural heritage built in stone. Consequently, it is of significance to identify the deterioration and develop appropriate restoration interventions. To this end, many researchers have classified the types of deteriorations that occur in stones [30–32]. In this study, the definitions recommended in ICOMOS-ISCS [32] were used in the detection of deteriorations in Gümüşler Monastery. In addition, the descriptions of the discontinuities in the monument and fresco losses that occur indoors, which are not included in the abovementioned dictionary, are presented in the following.

- Cracks: Non-planar and non-continuous deterioration form that develops in rocks in the presence of atmospheric processes and decrease and/or increase in the structural loads [22] (Fig. 4a).
- Discontinuities: Continuous and planar structures that develop in rocks owing to geological factors (Fig. 4b).
- Contour scaling: Separation of the upper surface in layers having different thicknesses (order of millimetres to centimetres) with the weakening of the outer surface of the rock owing to atmospheric effects [33] (Fig. 4c).
- Missing parts: Loss of fragments owing to the weakening of the engineering properties of the stones forming the monuments, due to atmospheric or biological processes [32] (Fig. 4d and e).
- Biological colonization: Because of the decomposition of the rock, the surface roughness increases over time and organic matter accumulates on the surfaces. Biological formations (lichen, fungus, and moss) develop in the presence of sufficient organic material and suitable humidity and temperature conditions [34] (Fig. 4f).
- Presence of higher plants: Rooted plants develop when a sufficient amount of organic material accumulates in the gaps, cracks, and discontinuities caused by atmospheric or anthropogenic effects in rocks [34] (Fig. 4g).
- Deposit: Accumulation of exogenic materials such as atmospheric dust, restoration residues, lime scale, and soot on the stone surface [32] (Fig. 4h).
- Efflorescence: The salts dissolved in water penetrate the rock through the capillary and/or percolation effect. Later, with the increase in the ambient temperature, the water evaporates and efflorescence occurs with the accumulation of salt crystals on the rock surface (Fig. 4i).
- Loss of fresco: Deterioration of some or all of the fresco created on the rock surface because of anthropogenic and atmospheric effects (Fig. 4j).

3. Methodology

With the technological developments in recent years, artificial-intelligence-based applications are rapidly substituting human-based applications in various domains, such as face recognition [35], natural language processing [36], and autonomous driving [37], to minimize human effort and errors. The development of DL methods, which represent artificial intelligence algorithms, has been inspired by the processing–perception process in the human brain. Based on these methods, LeCun et al. [38] created CNNs, which can capture and classify the features in images by processing the images in different layers. However, because the objects in the selected image have different positions and dimensions, extensive region selection must be performed, which makes the calculation highly complex. To overcome these problems Girshick et al. [39], a region-based object detection algorithm, R–CNN (regions with convolutional neural networks) has been developed. However, because this method must be applied separately for each region, the training and testing processes are highly time consuming. To solve this problem Girshick [40] proposed the Fast R–CNN technique, and Ren et al. [41] further enhanced the process speed by using the Faster R–CNN technique. Notably, using the Faster R–CNN, object detection applications can be implemented with a high speed and precision. In this method, the object boundaries are determined through a bounding box. However, when the number of objects in the image is high, the output is quite complex, which renders the interpretation difficult. He et al. [42] enhanced the Faster R–CNN by adding an extra

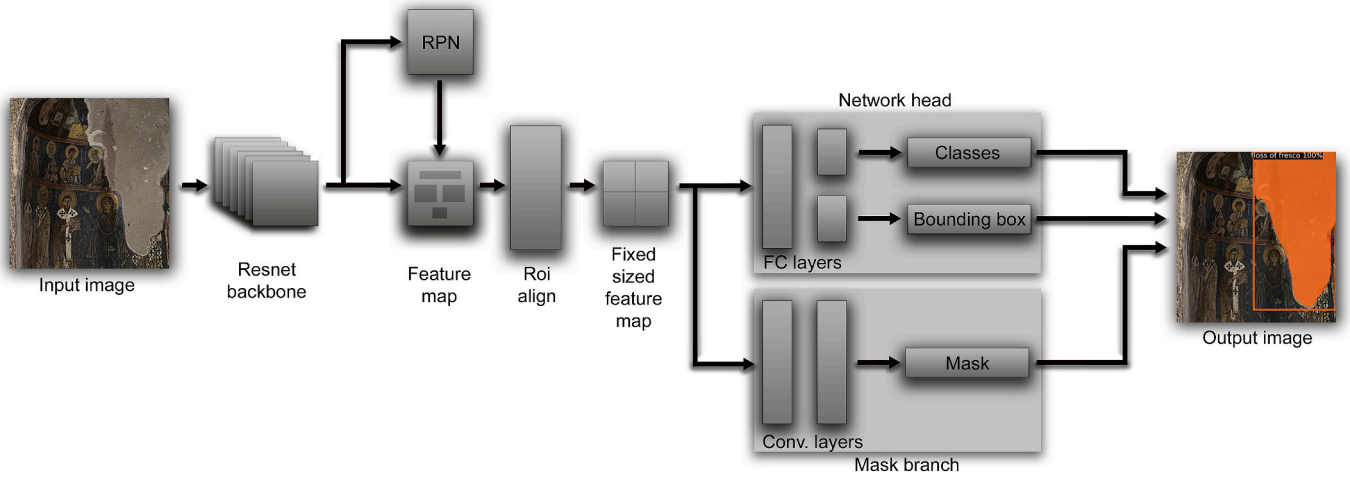

Fig. 5. The structure of Mask R–CNN Algorithm.

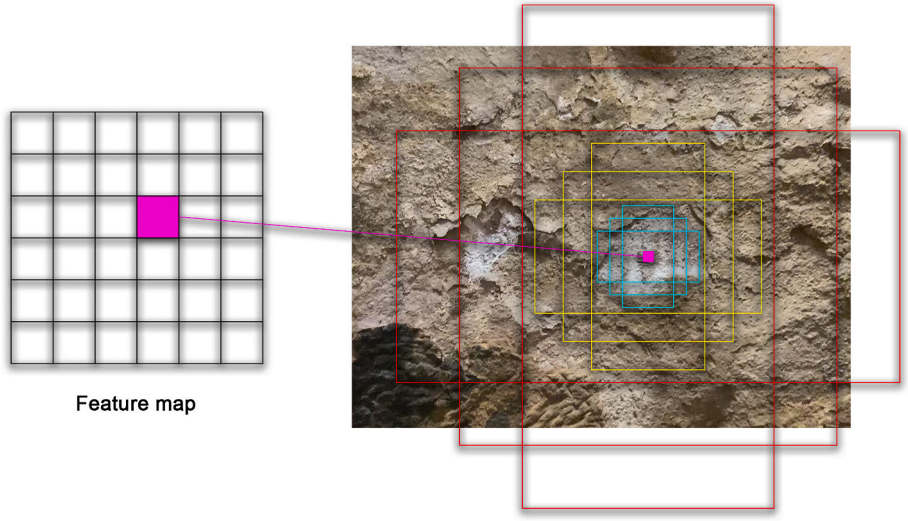

Fig. 6. Representation of the method's approach to nine anchors on the image.

branch to the R–CNN. This method can detect objects in complex images with a high sensitivity and create a quality segmentation mask for each object.

In the Mask R–CNN, end-to-end testing and training is implemented by integrating the region offers and mask application algorithms with CNNs. In this manner, the testing and training efficiency exponentially increases and the process mimics real perception. The process flow of the Mask R–CNN algorithm adopted in this study is shown in Fig. 5. First, the image is transferred to the backbone network. The first layers of this network perceive the features with low levels such as edges and corners, and the later layers perceive more complex features (such as cars, humans, and deterioration types). Neural networks with different depths can be designed to achieve a high accuracy in the backbone network. However, the higher accuracy rate may be achieved at the cost of the processing speed. In this study, the ResNet-101 model, which optimizes even in the cases of residual value feed is zero, was used to achieve a high accuracy within a reasonable training period. The feature maps obtained from the ResNet-101 are sent to the region proposal network (RPN) to determine the region offers. The RPN uses an anchor approach that formulates rectangular boxes with a specific aspect ratio on the image. Each point on the feature map is mapped to the original image with nine links (Fig. 6). As shown in the figure, the number of anchors for the feature map with dimensions of 6×6 is $6 \times 6 \times 9 = 324$. In addition, the anchors of the classification branch and border regression branch in the RPN are determined. Through this characteristic of the RPN, the object ratio differences in images acquired from different angles and distances can be eliminated. Subsequently, ROI align operation is implemented to ensure that all recommendations have the same dimensions. Finally, the sample is passed through the fully connected layer to determine the categories and coordinates of the proposed objects. The boxes created for the classes in this layer are grouped with the bounding box, and the highest estimate is selected. In the last stage, mask application is performed to the groups suggested for the object in the mask branch.

3.1. Implementation of mask R–CNN

Deep learning algorithms are an artificial intelligence approach that produces predictions by examining millions of data points between inputs and outputs. However, this method has some limitations. Sufficient number of images of the object groups to be detected primarily should be available in the training data set. In addition, images with different object exposures (such as shooting angle, distance, light and shadow values) should be used in this data set. Otherwise, the algorithm may fail. In order to overcome these limitations, the workflow scheme of the proposed method in the study is given in Fig. 7. In the data preparation

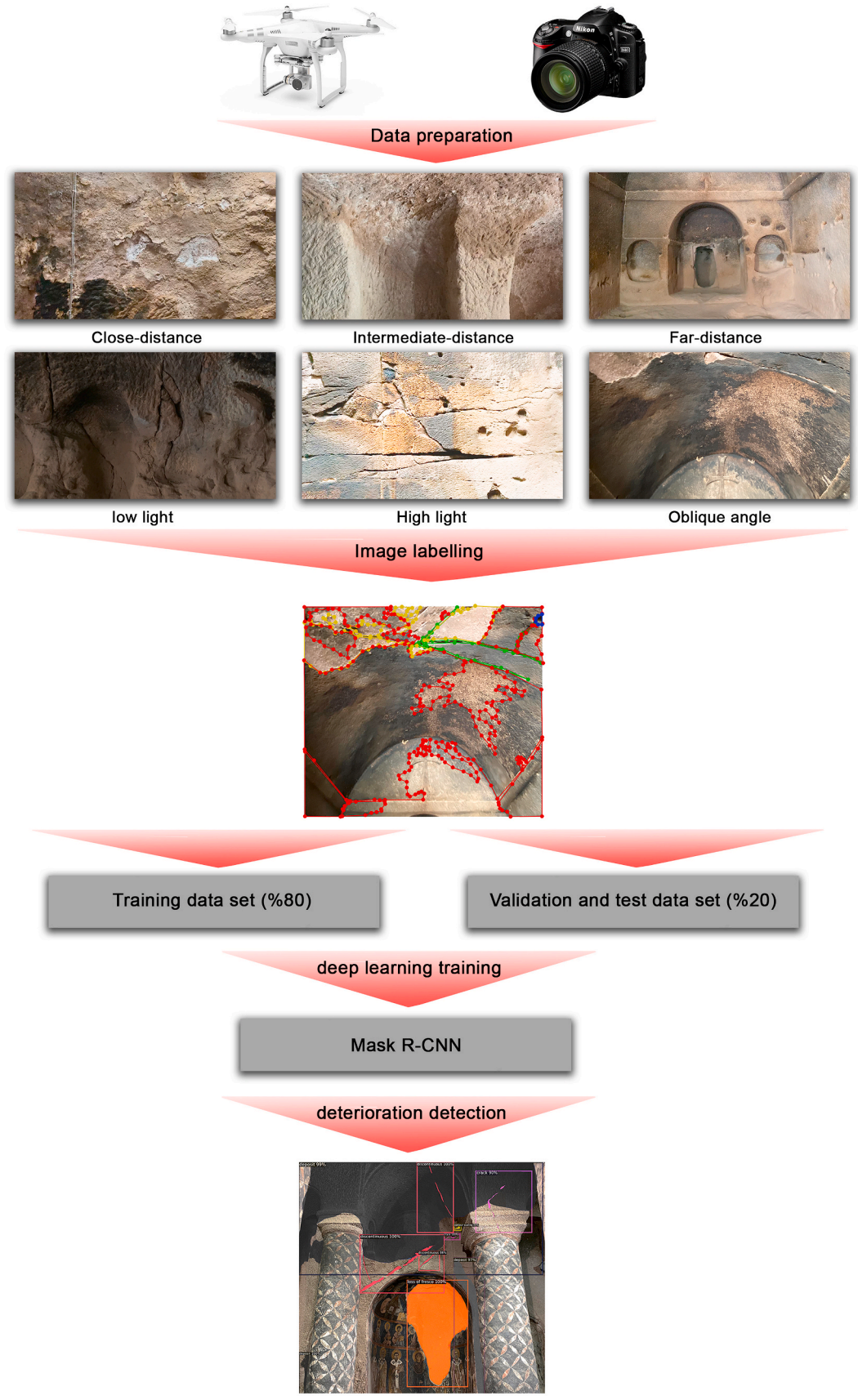

Fig. 7. Workflow chart of the proposed model.

Table 1
Labeling numbers of deterioration and loss of fresco.

Class	Number of labeling
biological colonization	2184
deposit	2013
crack	1879
discontinuous	1843
contour scaling	1692
higher plant	1439
missing part	1273
efflorescence	936
loss of fresco	126
Total	**13,385**

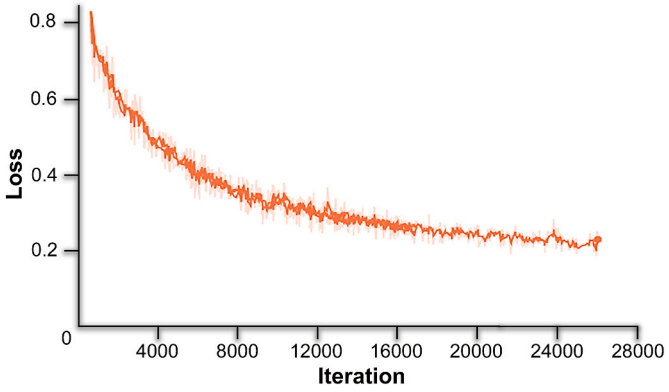

Fig. 8. Total loss curve for model.

stage in Fig. 7, a DJI Phantom 3 Advanced drone with a resolution of 12 megapixel (MP) and a digital camera were used to capture images. The images used for the proposed algorithm are collected from Gümüşler Monastery and various underground structures. In field studies, 1740 images with a resolution and dimensions of 72 dpi and 4000 × 3000, respectively, were compiled. In addition, images acquired at different distances (from 5 cm to 20 m) were obtained to ensure that the proposed model could detect different types of deterioration in indoor and outdoor spaces. In general, differences in the perspective between vertical and oblique images can lead to misjudgement regarding the appearance of deterioration type. Consequently, images with different perspectives were also used to train the model. In addition, due to the limited number of architectural decorative elements and frescoes in the monument, videos were compiled for the "missing parts" and "loss of frescoes". The frames in the obtained videos were recorded in the jpeg format to increase the number of images to ensure the intensive training of the algorithm. To train the model, all the images were converted to have dimensions of 800 × 600 px to reduce the training time. For the image labelling stage in Fig. 7, the coordinates and groups of the objects in the images were labelled, with 13,385 labels, via the open source LabelMe framework (Table 1). Notably, 80% (1392) of the 1740 images used in labelling were used in the training dataset, and the remaining 20% images were used in the validation and test sets. In the last stage before the training, all the labelled images were converted to json format.

Moreover, for the deep learning training stage in Fig. 7, Mask R–CNN was implemented in the TensorFlow framework. The training parameters, specifically, the learning rate, batch size, and momentum coefficient were set as 0.003, 16, and 0.9, respectively. The number of iterations expressing the number of weight updates of the model was determined to be 26,000.

In the test phase, two orthophotographs, two interior images, and two differently angled outdoor images were used to evaluate the performance of the model in conditions involving different distances and angles in indoor and outdoor spaces. The orthophotos of the courtyard facades of the monastery were obtained through the PhotoScan programme. A computer with a graphics processing unit (GPU) processor and NVidia GeForce GTX 1060 GPU graphics card was chosen to reduce the model training, testing, and validation periods.

4. Experimental results

The loss graph prepared to determine the estimation reliability of the proposed model is shown in Fig. 8. The loss value approaches 0.2 in 26,000 iteration steps. A value closer to zero indicates a higher similarity with the actual value, and thus, the estimation reliability of the algorithm can be considered to be reasonably high.

In addition to the average precision (AP) values, which indicate the percentage accuracy of the determination of the deterioration classes by using the proposed model, the mAP percentage, which is the average of these values, are presented in Table 2. The mAP of the model for 26,000 iterations is 98.186%, and the total training time is 268 min. In addition, an AP of 100% is achieved in determining cracks, contour scaling, biological colonization, presence of higher plants, deposits, and efflorescence. The AP values for the other types vary between 98.194% and 91.591%. The high iteration, AP and mAP values obtained in the training of the model were evaluated in the test phase on 2 orthophotographs prepared for the areas with the highest deterioration, 2 long-distance images with inclined shooting angles and 2 images of the interior. The orthophotograph of the northwest facade of the courtyard is shown in Fig. 9a, and the test result of the model is shown in Fig. 9b. As shown in Fig. 9b, the most prominent deterioration types on the facade correspond to deposits, discontinuities, and missing parts. Deposits are identified in the leaching zone in which the washing effect is low. Discontinuities, which cut across the courtyard facades in certain directions, corresponded to the most notable damage. Consequently, missing part formations were triggered over time in the lower levels of the pseudo plasters on the front and in the pseudo arches. Missing parts represent one of the most difficult deterioration types to be detected in rock carved monuments, primarily because of the lack of knowledge regarding the original shape or texture of the monument due to the re-exposure of the original rock unit that occurs after the loss of fragments. Nevertheless, the developed model can overcome this difficulty and determine the missing parts with a high accuracy (89–100%). Certain missing parts on the front could not be determined owing to the limited number of data in the training set. The test results for the northwest front indicate that the model can detect complex discontinuous formations with an accuracy of 83–100%. The accuracy rates for the other deterioration types, specifically, deposits, contour scaling, higher plants, and cracks are 97–100%, 79–100%, 97–100%, and 100%, respectively.

The orthophotograph of the north-eastern facade of the monument courtyard is shown in Fig. 10a. The most prominent deteriorations correspond to the missing parts (in the lifetime of the structure) as well as deposits and biological colonization (due to infiltration of surface water). The results obtained by the model for this facade are shown in

Table 2
Mean average precision (mAP), average precision (AP) and time values of the developed model.

Iteration	mAP (%)	AP (%) for contour scaling	AP (%) for biological colonization	AP (%) for higher plant	AP (%) for deposit	AP (%) for efflorescence	AP (%) for loss of fresco	AP (%) for discontinuous	AP (%) for crack	AP (%) for missing part	Time
26,000	98.186	100	100	100	100	100	98.194	96.314	94.782	91.591	4:28:21

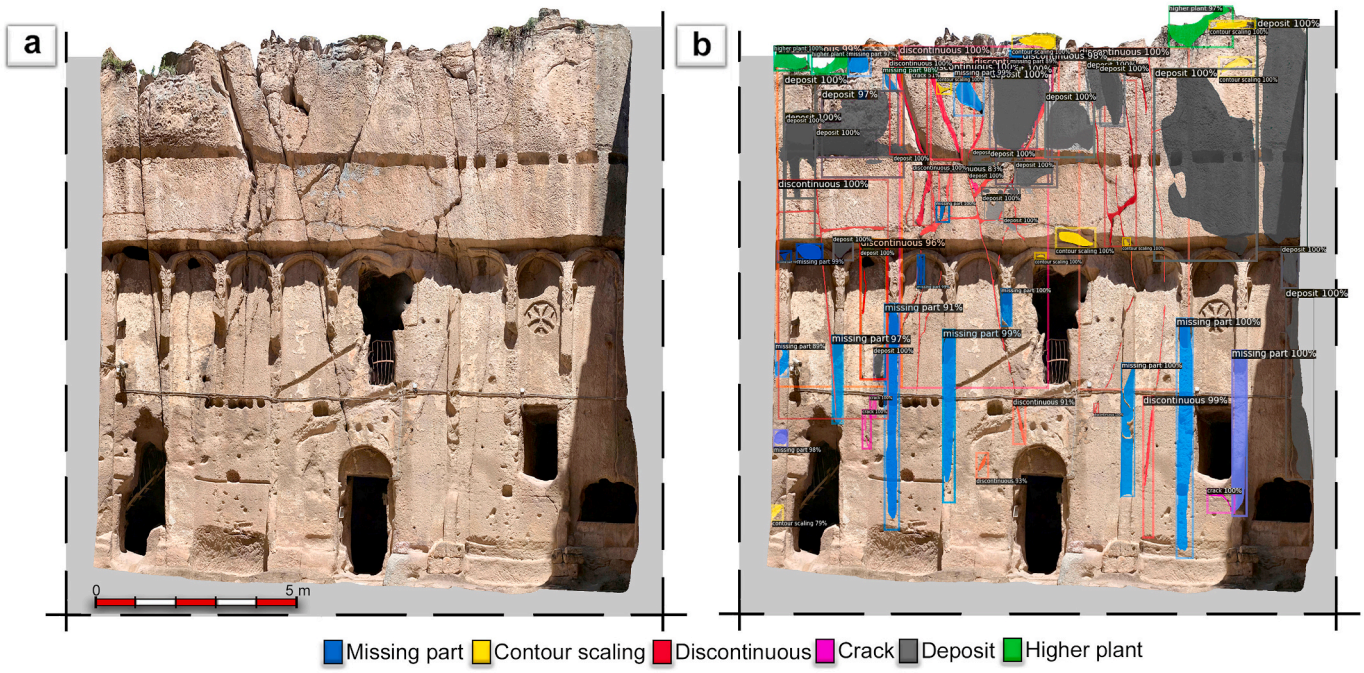

Fig. 9. Images of the northwest facade in the courtyard of the monument; a) orthophographic image, b) test results of model.

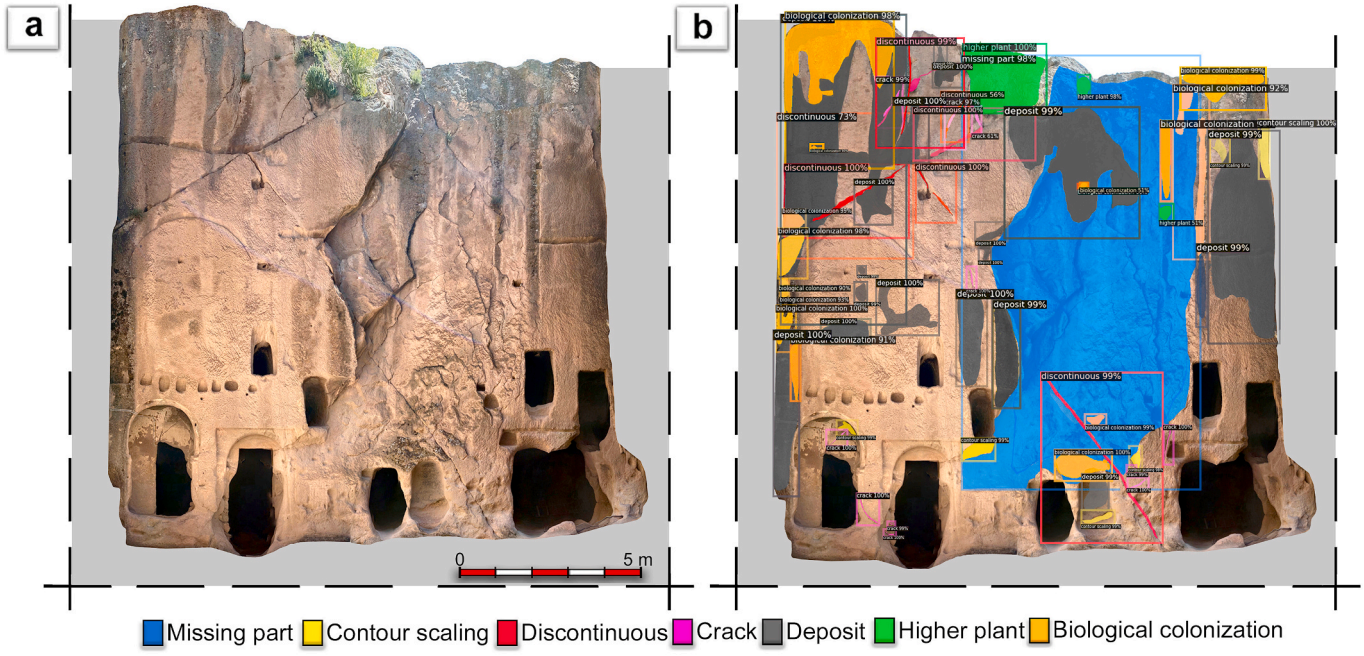

Fig. 10. Images of the northeastern facade in the courtyard of the monument; a) orthophographic image, b) test results of model.

Fig. 10b. The missing part spread over a wide area can be identified with an accuracy of 98%. The accuracy rates of other commonly observed deterioration types (biological colonization and deposits) in this facade are 86–100% and >99%, respectively. The relatively fewer cracks, contour scaling, and discontinuous type deteriorations on this facade can be determined with an accuracy of 97–100%, 98–100%, 99–100%, respectively.

To evaluate the performance of the model in conditions involving long-distance images obtained at wide and oblique angles, the images shown in Figs. 11a and 12a were used. These images belonged to different perspectives from the upper part of the monument. Soil accumulation occurred in the upper zone of the structure owing to the deterioration of the rock and transport of organic material by wind. Owing to the sufficient content of this organic material, biological colonization and higher plant formations occurred. In addition, deposits and biological colonization occurred in the infiltration zone owing to precipitation. The test results of both images are shown in Figs. 11b and 12b. According to Fig. 11b, the most accurate detection rates correspond to the presence of higher plants (88–100%) and deposits (87–100%). The corresponding rates for other types, such as missing parts, contour scaling, discontinuities, cracks, and biological colonization, are 60–100%. According to Fig. 12b, which corresponds to oblique angle

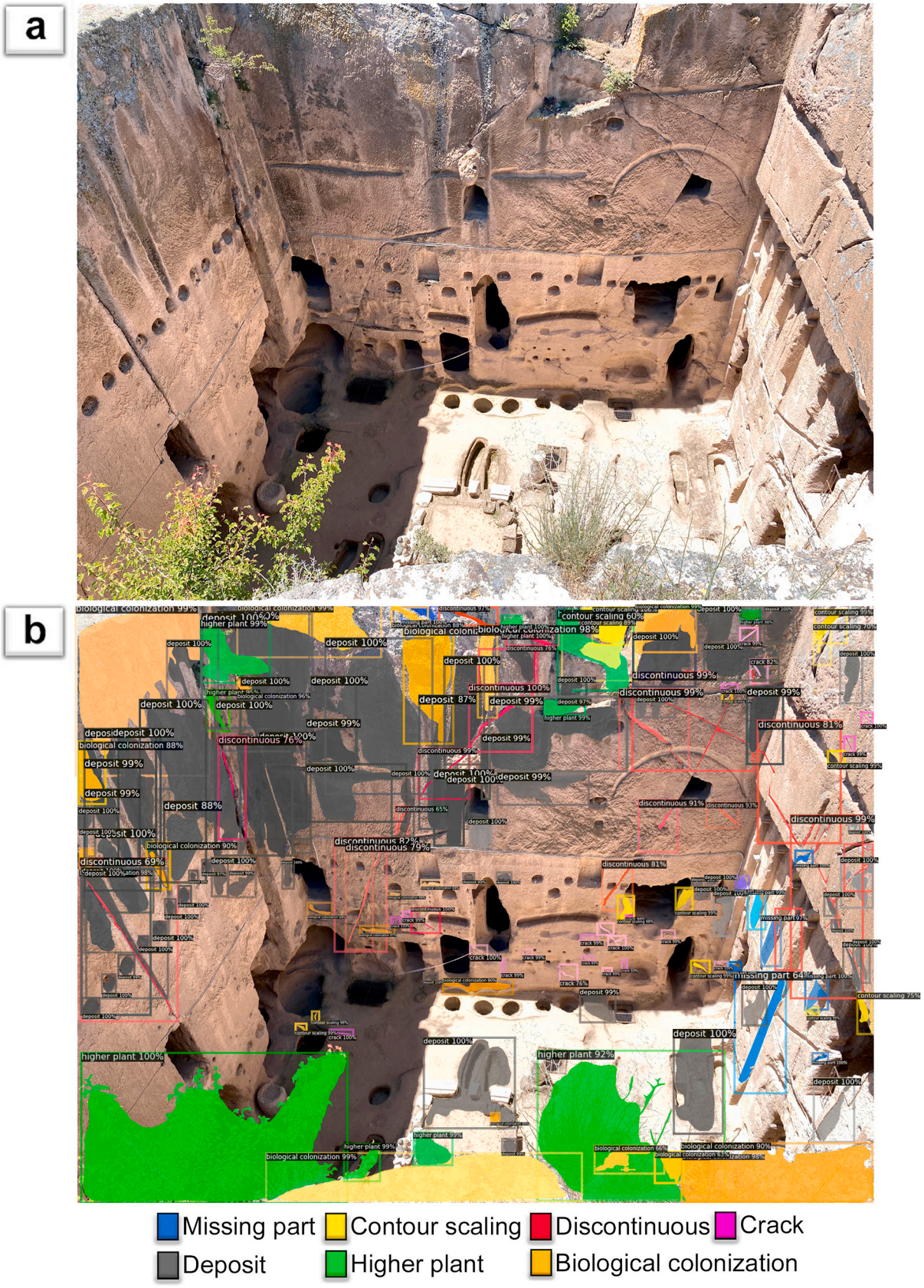

Fig. 11. View from northeast to southwest over the monument; a) original image, b) test results of model.

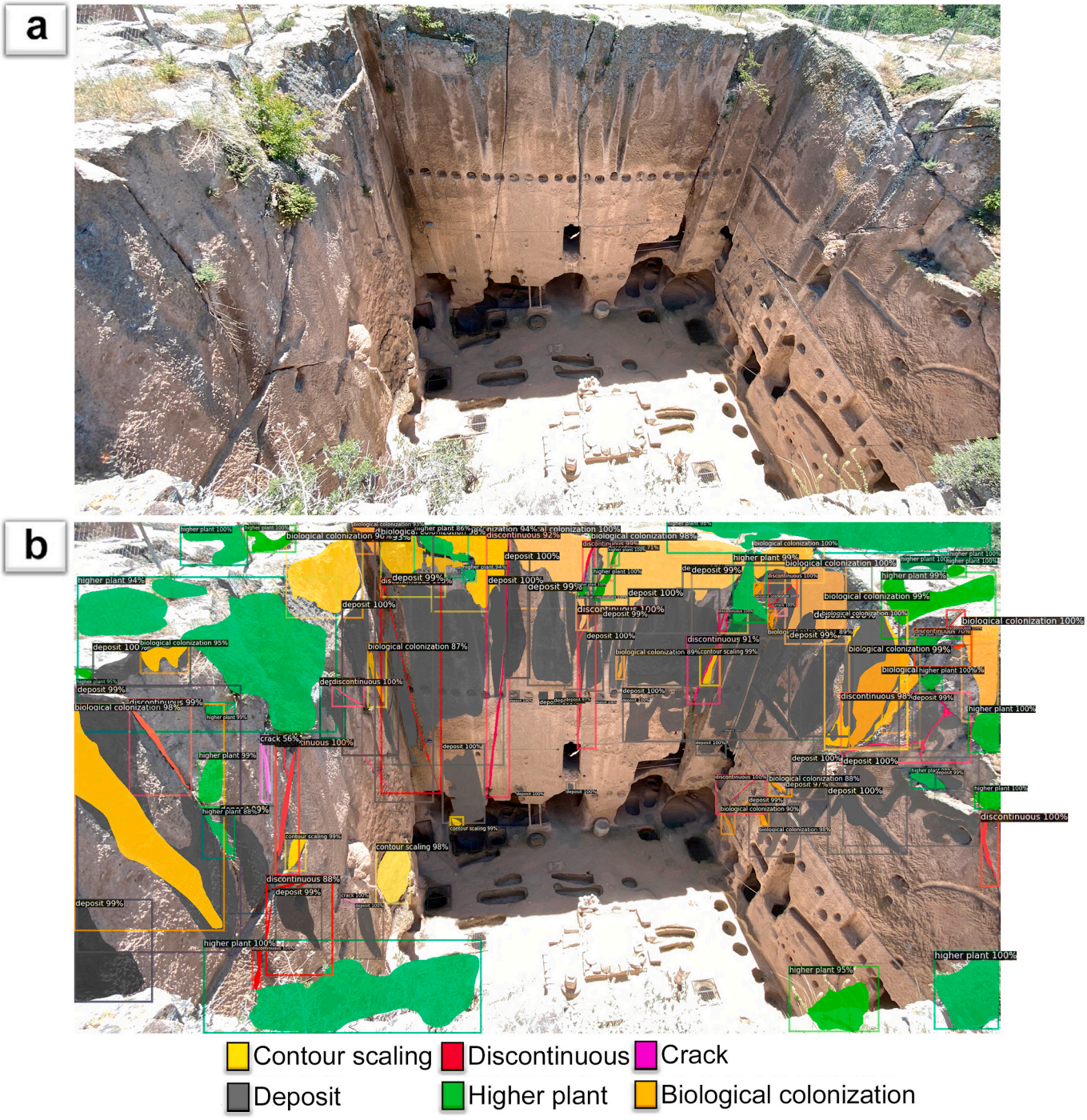

Fig. 12. View from northwest to southeast over the monument; a) original image, b) test results of model.

images, the cracks located next to the discontinuities can be determined with an accuracy of 56%. The accuracy rates to detect the deposits, contour scaling, presence of higher plants, biological colonization, and discontinuities are 97–100%, 98–99%, 88–100%, 87–100%, 70–100%, respectively.

The images shown in Figs. 13a and 14a were used to assess the performance of the model in the case of different types of deterioration in the interior sections. According to Fig. 13a, soot-induced deposit formation due to anthropogenic effects occurred on the interior walls and ceiling, and efflorescence occurred due to the increased concentration of salt minerals in water rising through percolation and capillary effects. The test results for this image, obtained using the developed algorithm, are shown in Fig. 13b. Deposits, one of the commonly occurring deteriorations in the interior sections, can be detected with an accuracy of 91–100%. Efflorescence is determined with an accuracy of 75%–99%. The detection rates of the limited contour scaling and cracks are 99–100% and 91–100%, respectively. In addition, the corresponding rate for discontinuities is 96%. According to Fig. 14a, discontinuities existed in the church wall and ceiling, owing to which, the surface water flowing from the surface to this section led to a loss of integrity of the frescoes on the wall. The corresponding results demonstrate that the fresco loss on the wall can be determined with a 100% accuracy (Fig. 14b). The other deterioration types detected in Fig. 14b correspond to the following accuracy rates: contour scaling (99%), deposits

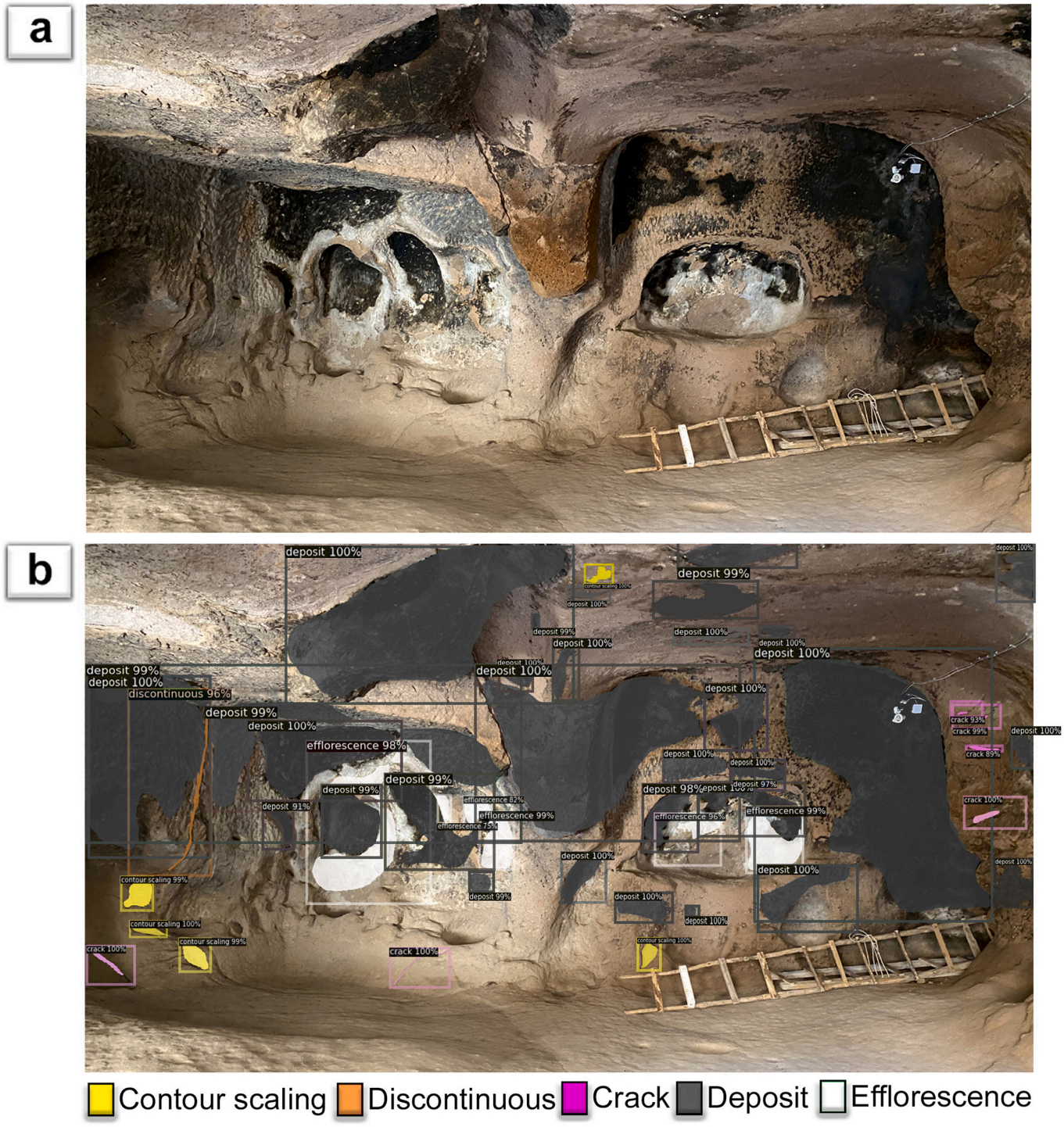

Fig. 13. Interior view of various places and outbuildings; a) original image, b) test results of model.

(97–99%), discontinuities (86–100%) and cracks (90–100%).

5. Discussion

Only a few DL-based object recognition studies have focused on historical building (Table 3). In the existing studies, material deterioration was determined using the Faster R–CNN and Mask R–CNN algorithms. The lowest AP values (0.890 and 0.891) for material deterioration detection corresponding to the models of Wang et al. [20] and Hatır et al. [23], respectively. Moreover, certain studies that used the Faster R–CNN algorithm to detect the brick material deterioration [19,43] reported values of 0.999, 0.900, and 0.875, for the efflorescence, spalling, and damage categories, respectively. Wang et al. [20] used the Mask R–CNN algorithm to detect the damage in tile materials and achieved an AP of 0.975. The AP was 0.883 when the Faster R–CNN technique was used to realize missing component detection in tile materials [21].

The proposed model was trained over nine deterioration types that develop in stone material, and AP values between 0.916 and 1.000 were achieved in tests. Compared with other studies, although the number of considered deterioration types in this study was high, superior AP values were obtained. This finding demonstrates that object recognition

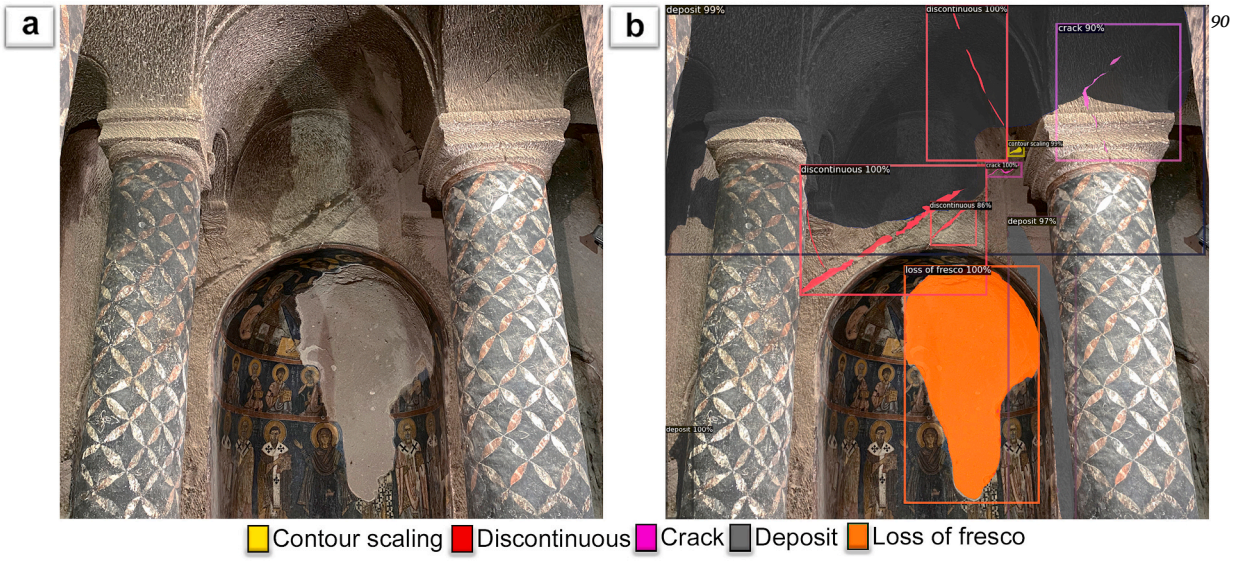

Fig. 14. Interior view of the church; a) original image, b) test results of model.

Table 3
Deep learning algorithms and AP rates developed specifically for cultural heritage.

Studies	Algorithm	Detected features	AP rate
Wang et al. [19]	Faster R–CNN	Efflorescence	0.999
		Spalling	0.900
Wang et al. [20]	Faster R–CNN	Pan tiles	0.890
		Roll roofing	0.910
	Mask R–CNN	Damage	0.975
Zou et al. [21]	Faster R–CNN	Missing component	0.883
Hatır and İnce [23]	Mask R–CNN	Stone lithologies	>0.891
Wang et al. [43]	Faster R–CNN	Damage	0.875

algorithms can realize successful recognition for more than one type of deterioration in cultural heritage. The additionally developed model exhibited a high detection performance in images with different light, shadow, and oblique angle conditions (Table 4). Notably, the success rate of the orthophotographs and indoor images with vertical angles was higher than that for the images with oblique angle. By increasing the number of images in the training set, the success rate for images pertaining to oblique angle can be increased. However, the DL technique was more effective in detecting the deterioration of large-scale monuments because it could overcome human errors that commonly occur in manual inspection.

6. Conclusion

Owing to the large spreading areas and heights of complex rock monuments, vision-based applications to examine the deterioration are time, cost, and labour intensive, and human-induced errors may occur. Computer-based applications can help overcome these problems. Therefore, in this study, an artificial-intelligence-based algorithm that could realize deterioration detection and mapping was developed in the context of the Gümüşler Monastery. The following key conclusions were derived:

The algorithm was trained over 26,000 iteration steps. The mAP value of the model was 98.186%, and the AP values for the deteriorations ranged between 91.591% and 100%.

A training set was created for each situation (different shooting angles, distances, and indoor/outdoor spaces) and all the deteriorations in the different images of the monument could be detected with high success rates.

The proposed model could automatically and accurately determine nine deterioration types with different colours, shapes, and textural features, ranging in size from millimetres to metres.

Although crack and discontinuities have similar forms, shapes, and colours, the proposed algorithm could distinguish these deterioration types caused by structural defects and geological factors.

The losses occurring in the pseudo plaster and pseudo arches of the structure could be determined to a notable extent, and this aspect demonstrates the feasibility of artificial-intelligence-based restitution studies in future applications.

This proposed model represents the first attempt to realize Mask R–CNN based deterioration detection and mapping in complex rock carved monuments. The ability of the algorithm in realizing automatic deterioration mapping with a reasonable training time and high accuracy can facilitate periodic examinations in conservation studies. The method can be applied to monuments containing similar types of deterioration in different regions. In monuments involving different types of deterioration, new images must be added to the data set to enhance the

Table 4
The accuracy rates of the developed model in different situations.

Case		Deterioration types/accuracy rate (%)								
	Figure number	Missing part	Contour scaling	Discontinuous	Crack	Deposit	Higher plant	Biological colonization	Loss of fresco	Efflorescence
Orthophograph	Fig. X	89–100	79–100	83–100	100	97–100	97–100	–	–	–
	Fig. X	98	98–100	99–100	97–100	99–100	51–100	86–100	–	–
Oblique angle	Fig. X	64–100	60–99	65–100	76–100	87–100	88–100	63–99	–	–
	Fig. X	–	98–99	70–100	56	97–100	88–100	87–100	–	–
Interior image	Fig. X	–	99–100	96	89–100	91–100	–	–	–	75–99
	Fig. X	–	99	86–100	90–100	97–99	–	–	100	–

model applicability. In addition, the deteriorations can be detected in real time by using cameras to monitor the monuments.

In addition to deteriorations determinations, in future researches, the period, style, building material and components of architectural or archaeological artefacts can be determined with DL. In this manner, DL-based algorithms can facilitate the preservation of cultural heritage for future generations.

Author statement

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Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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